MA1522 Linear Algebra for Computing
L2 Lesson 10



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Semester 1, 2026/27

Basis

Let V be a subspace of $\mathbb{R}^n$. A subset B of V is a **basis** for V if and only if

- B is linearly independent;
- $\text{span}(B) = V$.

Question 1

Find a basis for the solution set V of the homogeneous linear system

$$\begin{cases} x_1 + x_3 + x_4 - x_5 = 0 \\ x_2 + x_3 + 2x_4 + x_5 = 0 \\ x_1 + x_2 + 2x_3 + x_4 - 2x_5 = 0 \end{cases}.$$

Question 1: Solution

We first solve the linear system: $\begin{pmatrix} 1 & 0 & 1 & 1 & -1 \\ 0 & 1 & 1 & 2 & 1 \\ 1 & 1 & 2 & 1 & -2 \end{pmatrix} \xrightarrow[\text{Jordan}]{\text{Gauss-}} \begin{pmatrix} 1 & 0 & 1 & 0 & -2 \\ 0 & 1 & 1 & 0 & -1 \\ 0 & 0 & 0 & 1 & 1 \end{pmatrix}$.

Thus $x_3 = s$, $x_5 = t$, $x_1 = -s + 2t$, $x_2 = -s + t$, $x_4 = -t$, i.e.

$$\begin{aligned}(x_1, x_2, x_3, x_4, x_5) &= (-s + 2t, -s + t, s, -t, t) \\ &= s(-1, -1, 1, 0, 0) + t(2, 1, 0, -1, 1),\end{aligned}$$

so that $V = \text{span}\{(-1, -1, 1, 0, 0), (2, 1, 0, -1, 1)\}$.

Easy to see that $\{(-1, -1, 1, 0, 0), (2, 1, 0, -1, 1)\}$ is linearly independent.

Answer: $\{(-1, -1, 1, 0, 0), (2, 1, 0, -1, 1)\}$.

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Finding a basis for $\mathrm{span}(S)$

Let $S = \{\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_k\} \subseteq \mathbb{R}^n$.
Two ways to find a basis for $V = \mathrm{span}(S)$:

1. **Treat all vectors as rows vectors.**

   Let $\boldsymbol{A} = \begin{pmatrix} \boldsymbol{v}_1 \\ \boldsymbol{v}_2 \\ \vdots \\ \boldsymbol{v}_k \end{pmatrix}$, and let $\boldsymbol{R} = \begin{pmatrix} \boldsymbol{r}_1 \\ \boldsymbol{r}_2 \\ \vdots \\ \boldsymbol{r}_k \end{pmatrix}$ be an REF of $\boldsymbol{A}$.

   Then the **nonzero** $\boldsymbol{r}_i$'s form a basis for V.

2. **Treat all vectors as columns vectors.**

   Let $\boldsymbol{A} = \begin{pmatrix} \boldsymbol{v}_1 & \boldsymbol{v}_2 & \cdots & \boldsymbol{v}_k \end{pmatrix}$, and let $\boldsymbol{R} = \begin{pmatrix} \boldsymbol{c}_1 & \boldsymbol{c}_2 & \cdots & \boldsymbol{c}_k \end{pmatrix}$ be an REF of $\boldsymbol{A}$.

   If $\boldsymbol{c}_{i_1}, \boldsymbol{c}_{i_2}, \ldots, \boldsymbol{c}_{i_r}$ are the pivot columns of $\boldsymbol{R}$, then $\{\boldsymbol{v}_{i_1}, \boldsymbol{v}_{i_2}, \ldots, \boldsymbol{v}_{i_r}\}$ is a basis for V.

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Solution.

1. $\boldsymbol{A} = \begin{pmatrix} 1 & 0 & -1 \\ -1 & 2 & 3 \\ 0 & 3 & 0 \\ 1 & -1 & 1 \end{pmatrix} \xrightarrow[\text{Elimination}]{\text{Gaussian}} \begin{pmatrix} 1 & 0 & -1 \\ 0 & 2 & 2 \\ 0 & 0 & -3 \\ 0 & 0 & 0 \end{pmatrix} = \boldsymbol{R}.$

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   Answer: $\{(1, 0, -1), (0, 2, 2), (0, 0, -3)\}$

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Find a basis for $V = \text{span}\{(1,0,-1),(-1,2,3),(0,3,0),(1,-1,1)\}$.

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Question 3

Let $S = \{\boldsymbol{v}_1, \boldsymbol{v}_2, \boldsymbol{v}_3, \boldsymbol{v}_4, \boldsymbol{v}_5\}$, where

$$\boldsymbol{v}_1 = (1, 0, 1, 3),\ \boldsymbol{v}_2 = (2, -1, 0, 1),\ \boldsymbol{v}_3 = (-1, 3, 5, 12),$$
$$\boldsymbol{v}_4 = (0, 1, 2, 5),\ \boldsymbol{v}_5 = (3, -1, 1, 4).$$

Let $V = \operatorname{span}(S)$.

a. Find a subset $S' \subseteq S$ such that S' is a basis for V.
b. Express the vectors in $S \setminus S'$ as linear combinations of vectors in S'.

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Solution.

$$\begin{pmatrix} \boldsymbol{v}_1 & \boldsymbol{v}_2 & \boldsymbol{v}_3 & \boldsymbol{v}_4 & \boldsymbol{v}_5 \end{pmatrix} = \begin{pmatrix} 1 & 2 & -1 & 0 & 3 \\ 0 & -1 & 3 & 1 & -1 \\ 1 & 0 & 5 & 2 & 1 \\ 3 & 1 & 12 & 5 & 4 \end{pmatrix} \xrightarrow[\text{Jordan}]{\text{Gauss-}} \begin{pmatrix} 1 & 0 & 5 & 2 & 1 \\ 0 & 1 & -3 & -1 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

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Thus $S' = \{\boldsymbol{v}_1, \boldsymbol{v}_2\}$ and

$$\boldsymbol{v}_3 = 5\boldsymbol{v}_1 - 3\boldsymbol{v}_2,\ \boldsymbol{v}_4 = 2\boldsymbol{v}_1 - \boldsymbol{v}_2,\ \boldsymbol{v}_5 = \boldsymbol{v}_1 + \boldsymbol{v}_2.$$

□

Basis for $\mathbb{R}^n$

Theorem

Let $S \subseteq \mathbb{R}^n$.

1. *If $|S| \neq n$, then S is* **not** *a basis for $\mathbb{R}^n$.*
2. *Suppose that $S = \{\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_n\}$, and let $\boldsymbol{A} = \begin{pmatrix} \boldsymbol{v}_1 & \boldsymbol{v}_2 & \cdots & \boldsymbol{v}_n \end{pmatrix}$. The following statements are equivalent:*
   a. *S is a basis for $\mathbb{R}^n$.*
   b. $\operatorname{span}(S) = \mathbb{R}^n$.
   c. *S is linearly independent.*
   d. *$\boldsymbol{A}$ is invertible.*
   e. $\operatorname{RREF}(\boldsymbol{A}) = \boldsymbol{I}_n$.
   f. $\det(\boldsymbol{A}) \neq 0$.

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Question 4

Let $V = \text{span}\{(1,1,0,0),(1,0,0,1),(0,1,1,0),(0,0,1,1)\}$.

Find disjoint subsets $B, C \subseteq \mathbb{R}^4$ such that B is a basis for V and $B \cup C$ is a basis for $\mathbb{R}^4$.

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Let $B = \{(1,1,0,0),(0,-1,1,0),(0,0,1,1)\}$ and $C = \{(0,0,0,1)\}$. Then B is a basis for V, and $B \cup C$ is a basis for $\mathbb{R}^4$. □

Next Lesson ...

We will explore **Sections 3.7 and 3.8**.
Please watch/review the video clips for these sections before then.